



SYDE252 - W01A - Course Intro

Dr. Matt Borland

Activity

5 minutes.

Work in teams of 4.

1. Come up with a definition of a signal and examples of signals.
2. Come up with a definition of a system and examples of systems.
3. Write down 2 - 3 things you've heard about this course.

Teaching Team

Instructor:

Prof. Matt Borland, PhD

mjborlan@uwaterloo.ca

Office: E7-6306

Teaching Assistants:

Niloofer Azad – tutorials and support.

Samyar Goodarzi – mostly marking.

Teaching Team

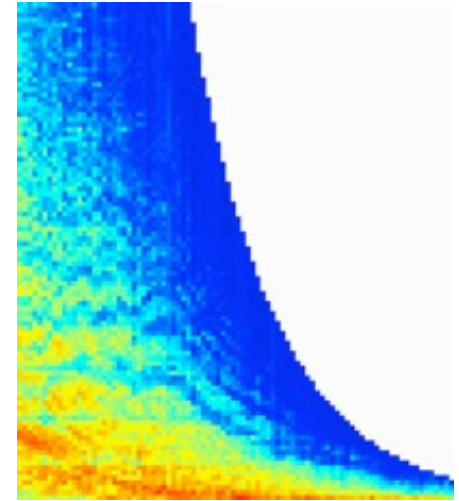
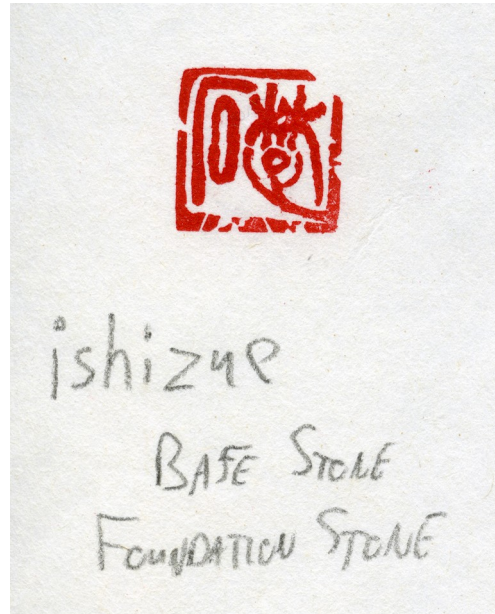
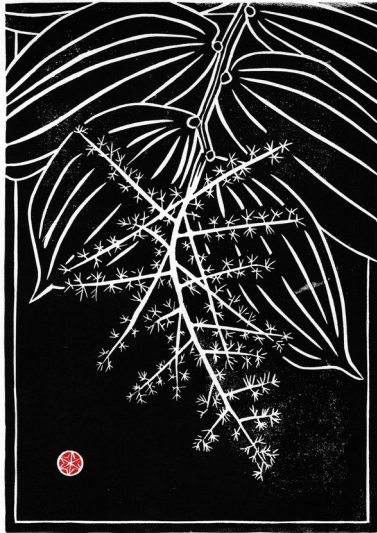
Contact:

We are always available to help you with the course, so feel free to contact us via email to set up a meeting outside of class time.

Office Hours:

Prof. Borland will be available for office hours on Mondays from 12:30-1:20pm in E7-6306.

Who is this guy?



Land



Syllabus

<https://outline.uwaterloo.ca/viewer/view/njmvtn>

Schedule

MTE252 S2026 Course Schedule -- schedule subject to change - changes will be communicated on UW-LEARN

NOTE: The three make-up lecture slots won't be used. Use them to review material or do assignments independently.

Lecture Monday 1:30pm-2:20pm E7-3353			Lecture Wednesday 1:30pm-2:20pm E7-3353		Tutorial Friday 9:30pm-10:20am E7-3353		Lecture Friday 1:30pm-2:20pm E7-3353		Readings - Linear Systems and Signals 3rd Edition - Lathi and Green	Assignments
W1	11-May	Intro, Course overview, Expectations, What is a Signal?	13-May	Basic Signal Properties	15-May	Imaginary and Complex Numbers - Review	15-May	Signal Manipulation	Ch. 8 as required, Ch. 1: 1.1-1.8, 1.12	A1
W2	18-May	HOLIDAY	20-May	Complex Signals and Unit Impulse	22-May	Linearity and Time Invariance NOTE - LECTURE AND TUTORIAL SWAPPED. THIS IS A LECTURE.	22-May	Signal Properties + Manipulation NOTE - LECTURE AND TUTORIAL SWAPPED. THIS IS A TUTORIAL.		
W3	25-May	Convolution	27-May	LTI Systems	29-May	Convolution	29-May	System Properties		
W4	01-Jun	System Response	03-Jun	Characteristic Equations	05-Jun	Quiz Review + Questions - must send in advance to TA	05-Jun	Quiz 1 - W1-3	Ch. 2: 2.1-2.6, 2.8, 2.9	A2
W5	08-Jun	Zero Input Response	10-Jun	Unit Impulse Response	12-Jun	System Response	12-Jun	Total Response		
W6	15-Jun	System Stability	17-Jun	System Response Review	19-Jun	Midterm Review - send questions in advance	19-Jun	System Response Review		
W7	22-Jun	MIDTERM WEEK - Lectures Cancelled					THURSDAY JUNE 25	Midterm - 9:30-11:30am - W1-6		
W8	29-Jun	DT Time Domain Analysis	01-Jul	HOLIDAY	03-Jul	Exponential Functions and Complex Numbers	03-Jul	Laplace Transform 1	Ch. 3: 3.1, 3.13, Ch 4: 4.1, 4.2	A3
W9	06-Jul	Laplace Transform 2	08-Jul	Laplace Transform 3	10-Jul	Laplace Transform	10-Jul	Laplace Transform 4	Ch. 4: 4.2, 4.3, 4.5, 4.8, 4.9, 4.13	A4
W10	13-Jul	Fourier Series	15-Jul	Fourier Transform 1	17-Jul	Laplace Transform and Fourier Series	17-Jul	Fourier Transform 2	Ch.6: 6.1, 6.2, 6.8, Ch. 7: 7.1-7.5	A5
W11	20-Jul	Fourier Transform 1	22-Jul	Fourier Transform 1	24-Jul	Quiz Review + Questions - must send in advance to TA	24-Jul	Quiz 2 - W8-10	Ch. 7: 7.1-7.5, 7.10	
W12	27-Jul	Sampling +DFT 1	29-Jul	Sampling +DFT 2	31-Jul	Project Help	31-Jul	Sampling +DFT 3	Ch. 8: 8.1-8.8, Ch. 9: 9.1-5, 9.8	Project Due
W13	TUESDAY AUG 4TH	Z-Transform + FFT	05-Aug	FFT + Course Wrap Up						

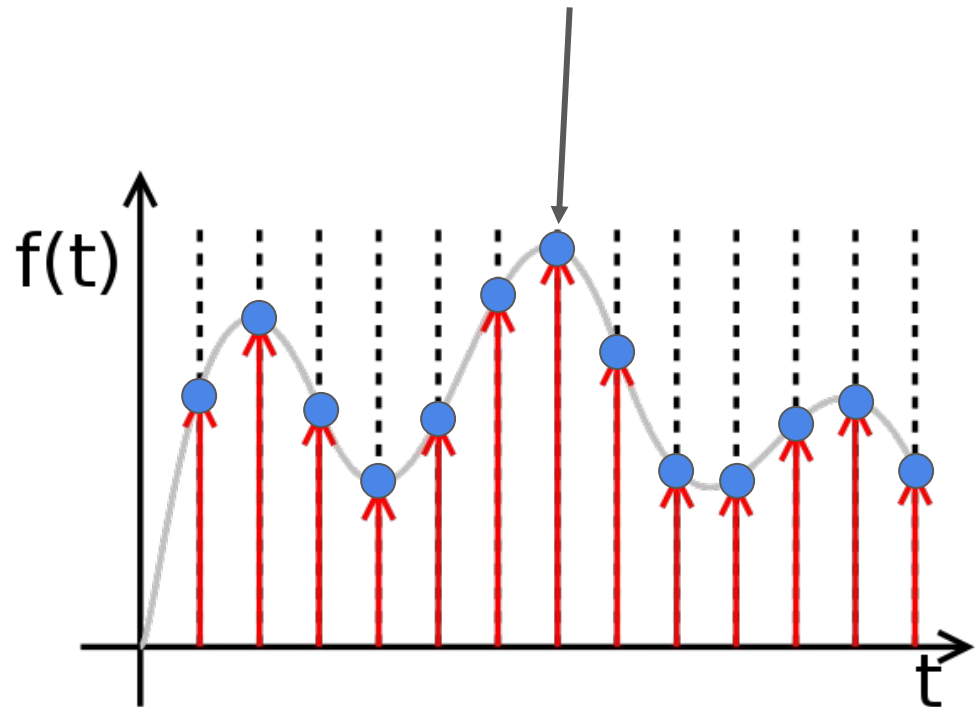
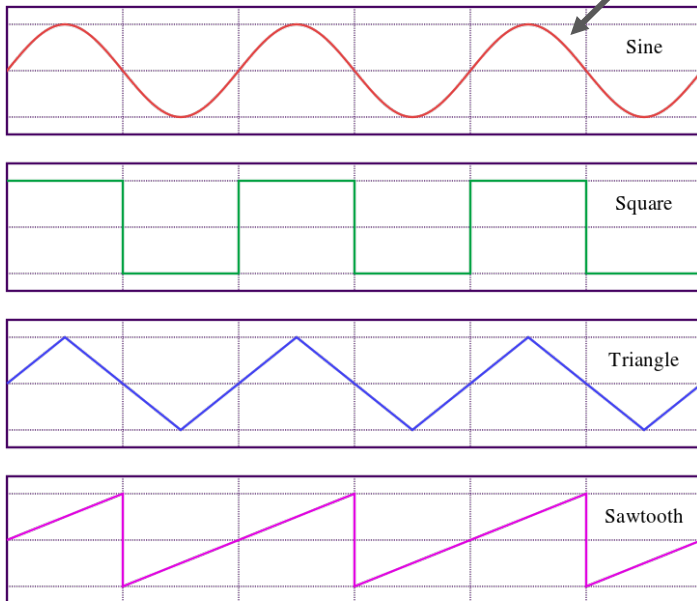
What's a Signal?

A **signal** as referred to in [communication systems](#), [signal processing](#), and [electrical engineering](#) is a function that "conveys [information](#) about the behavior or attributes of some phenomenon".^[1] In the physical world, any quantity exhibiting variation in time or variation in space (such as an image) is potentially a signal that might provide information on the status of a physical system, or convey a [message](#) between observers, among other possibilities.^[2] The *IEEE Transactions on Signal Processing* states that the term "signal" includes [audio](#), [video](#), speech, [image](#), communication, geophysical, sonar, radar, medical and musical signals.^[3]

Signal. (2017, April 23). In *Wikipedia, The Free Encyclopedia*. Retrieved 02:34, May 2, 2017, from

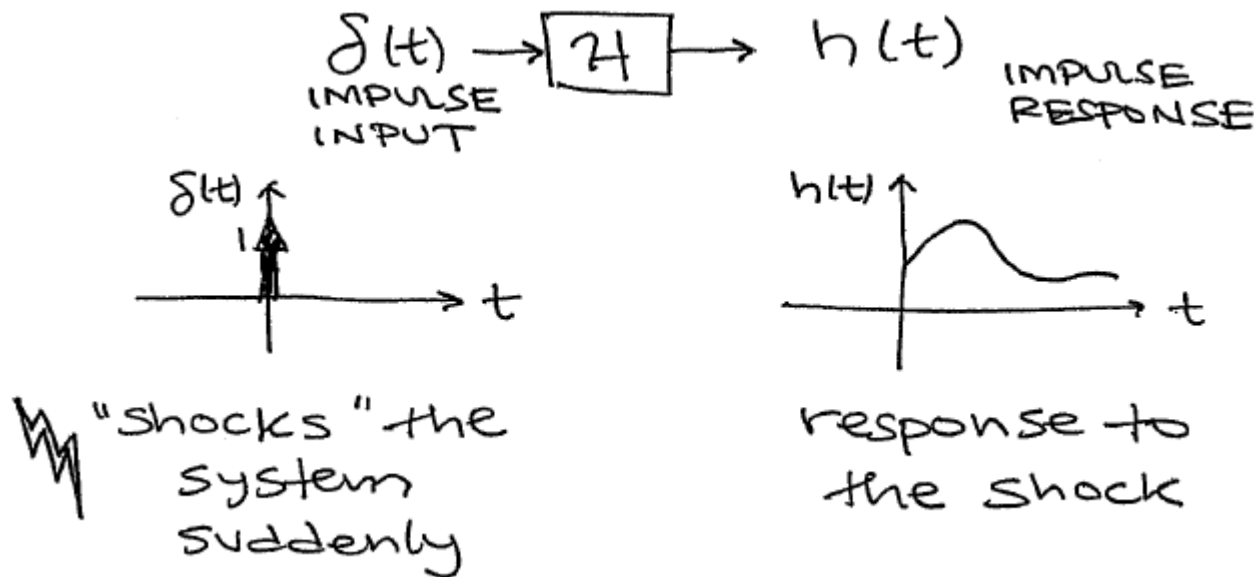
The most important things you need to learn!

Signals can be continuous or discrete.



The most important things you need to learn!

Systems (with special properties) can be characterized by their Impulse Response



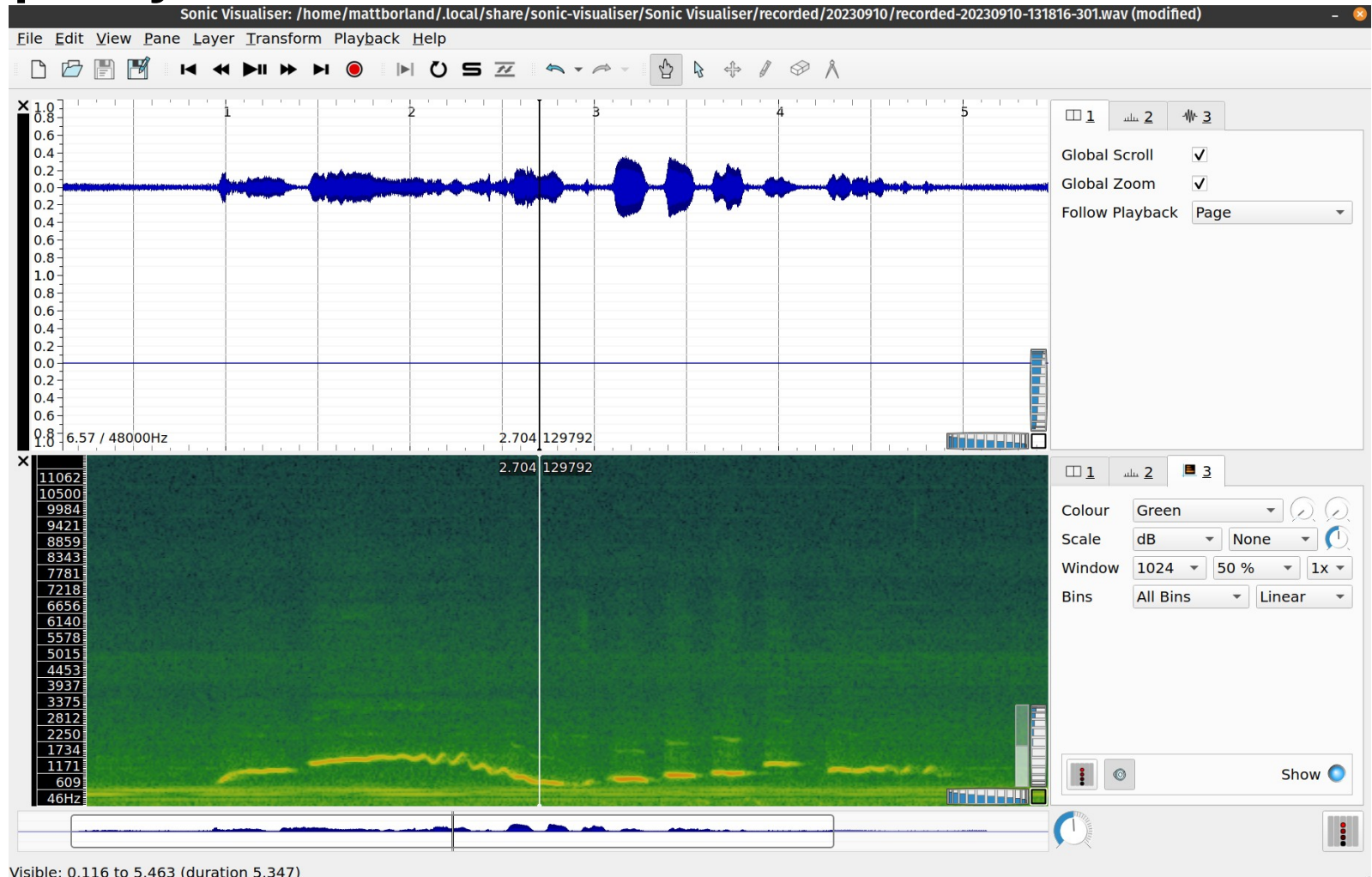
Drum demo...

Systems (with special properties) can be characterized by their Impulse Response

Download Sonic Visualizer
and try this for yourself!

<https://sonicvisualiser.org/>

Whistling a melody (with changing pitch (aka frequency))



The most important things you need to learn!

Signals can be represented in multiple domains, but they all are representing the same thing, the signal.



The most important things you need to learn!

Time Domain $\longleftrightarrow$ Frequency Domain.

